

AI Content Evaluation Engine

Tricky Python vs C++ Questions

Q1 — Python vs C++ Loops

Unlike C++, which has three loops (for, while, do-while), Python only has two — for and while — so how do you replicate do-while behavior in Python where the body must run at least once?

Answer:

Use while True: with the logic inside and a break at the end after checking the condition.

Q2 — Mutable Default Arguments

If a Python function is defined with a mutable default like `def f(lst=[]): lst.append(1); return lst`, what unexpected behavior occurs on repeated calls, and why doesn't this happen in C++?

Answer:

Python evaluates the default once at definition, so the same list is reused every call. C++ re-evaluates defaults each time.

Q3 — Integer Division with Negatives

C++ computes `-7 / 2` as `-3` (truncates toward zero), but Python computes `-7 // 2` as `-4` (floors toward negative infinity) — which behavior silently breaks logic when switching languages?

Answer:

Python's floor division breaks C++ assumptions; use `int(-7 / 2)` in Python to match C++ truncation behavior.

Q4 — Pass by Value vs Object Reference

In C++ you can pass by reference with `&`, but Python uses pass-by-object-reference — so does reassigning an integer inside a Python function affect the caller, and what about mutating a list?

Answer:

Reassigning an integer does not affect the caller (immutable), but mutating a list in-place does (mutable, same object).
